

Wai Yan Min Ko Ko

Singapore | +65 8607 0323 | waiyanminkoko@gmail.com | www.linkedin.com/in/waiyanminkoko/ | waiyanminkoko.github.io/portfolio/



EDUCATION

NANYANG TECHNOLOGICAL UNIVERSITY (NTU)

Bachelor of Engineering (Hons) in Computer Engineering with Minor in Entrepreneurship

SINGAPORE

August 2024 – December 2027

SINGAPORE POLYTECHNIC (SP)

Diploma with Merit in Mechatronics and Robotics (GPA: 3.90/4.00)

SINGAPORE

April 2021 – March 2024

Director's Honour Roll

Secretary & Community Program Organizer of the SP International Students' Club (SPISC)

Obtained a Professional Certificate in Computer Science for Artificial Intelligence (Harvard)

Bukit Panjang Government High School

Singapore-Cambridge GCE O-Level Graduate

SINGAPORE

January 2019 – November 2020

MOE ASEAN Scholar (Secondary 3 – Secondary 4)

SKILLS

Software and Programming

- C++ & C
- Python
- HTML & CSS
- JAVA
- React
- ROS
- Linux
- Inventor
- Mastercam
- AI App Dev.
- Canva
- Prusa
- AutoCAD
- Figma

Technical Skills

- Raspberry Pi
- Arduino
- Project Management
- Programmable Logic Controller (PLC)
- KUKA (Cobot)
- Networking
- 3D Printing
- Autonomous Mobile Robot Deployment

Core Skills

- Problem Solving
- Adaptability
- Self-Management
- Collaboration
- Team Management

PROJECT EXPERIENCE

Full-stack Internship Placement Management System (Web Development)

NTU

Project Head & Programmer

September 2025 – November 2025

- Constructed a full-stack, role-based Internship Placement Management System leveraging Java (Spring Boot) and React, implementing secure RESTful APIs, advanced reporting, and multi-level approval workflows for students, company representatives, and staff.
- Engineered robust CSV-based data persistence (project requirements) with thread-safe operations, enforced complex business rules (eligibility, application limits, placement logic), and delivered a responsive Material-UI frontend for real-time, user-friendly internship management.

Generative AI Lecture Companion Web App (AI Project Development)

NTU

Project Head & Programmer

March 2025

- Devised a full-stack AI Lecture Companion Web App utilising React for the final product and Streamlit for the prototype, enabling students to transcribe, analyse, and interact with lecture materials through audio processing and Retrieval-Augmented Generation (RAG) techniques.
- Integrated open-source AI models (Faster-Whisper for transcription and Llama3-70B via Groq API) to deliver intelligent Q&A, Key Concepts, Summarization, and Notes Taking features, enhancing the learning experience with seamless user interfaces and robust backend services.

Automated Guided Vehicle (Arduino & Fabrication)

SINGAPORE POLYTECHNIC

Project Head & Programmer, Awarded Top 6 Innovative Designs

October 2022 – February 2023

- Designed, fabricated, and programmed an Automated Guided Vehicle (AGV) with a specialised gripper capable of autonomous line-tracking and pick-and-place tasks to navigate complex maps and transport small payloads.
- Demonstrated expertise in utilising technologies, such as an Arduino Board and diverse output shields, to power and control the Automated Guided Vehicle (AGV). Designed various AGV components using Inventor, while programming the fabrication process through Mastercam, and subsequently programmed the AGV with C++.

Personal IOT Assistant Device (Raspberry Pi & Web Development)

SINGAPORE POLYTECHNIC

Project Head & Programmer

October 2022 – February 2023

- Steered a team of 2 in developing a personal IOT assistant device using a Raspberry Pi board emulating a smart home environment.
- Collaborated with team members to develop a functional Python program and web application using HTML and CSS, implementing best practices for coding and design, and troubleshooting technical issues to ensure efficient and maintainable code.

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PROFESSIONAL WORK EXPERIENCE

BOTSYNC Pte. Ltd.

SINGAPORE

Mechatronics & Robotics Intern (Operations, Deployment & R&D)

September 2023 – February 2024

(Key Skills: ROS, Linux, AMR Deployment, Prototyping, Client & Partner Collaboration, Project Management)

- Engineered comprehensive testing of cutting-edge autonomous navigation algorithms and workflow enhancement while cooperating with the software team to perform refinements, ensuring optimal performance.
- Spearheaded compelling robot demonstrations tailored to clients, showcasing the company's AMRs' capabilities. Driven by a commitment to results, these demonstrations contributed to increased sales and deeper client engagement.
- Directed and coordinated the seamless deployment of the AMRs at CATERPILLAR Logistics Warehouse & KEMIN Industries (Asia) Pte Ltd, ensuring the automation of the clients' work processes.
- Provided dedicated service and support for AMRs at Bollore Logistics (SG), Sanmina Corporation (SG), Wong Fong Engineering Works (SG), Mahindra Logistics (India), and Adyawinsa Group (Indonesia), ensuring seamless operation and client satisfaction.

Hwa Chong Institution Boarding School (HCIBS)

SINGAPORE

Head of the Office-Student Assistant Team (Estate & IT Department) (Part-time)

April 2021 – Present

(Key Skills: Web Automation Scripting, Web Design, Excel Automation, Mentorship & Training, Team Management)

- Mentor and lead a high-performing team of 10 NUS students, applying experience gained from training a diverse group of 21 student staff (NUS, SMU, ITE) on essential facilities and operational procedures to maintain consistent excellence in office support.
- Streamlined the procedure for the web developer to build and launch the website, saving the web developer 1 week of labour by creating a sitemap plan for the Frontend and Backend process of HCIBS internal Defect and Cleaning Report System.
- Developed an automated Student Attendance Management Tool using Python, Selenium, and Pandas to streamline data extraction from Excel and automate attendance marking on the HCIBS portal, reducing manual effort and improving data accuracy.

Technology and Trust (TNT)

YANGON, MYANMAR

Part-time Computer Technician & Sales Assistant

January 2012 – October 2018

(Key Skills: Problem-Solving, Decision Making, Customer Orientation, Communication)

- Assembled, fixed, and upgraded customers' computers and networks, boosting the processing speed of PCs, internet, and server connections, while ensuring mutual trust and strong business relationships.
- Consulted with clients to diagnose and resolve hardware and software pain points, driving a measurable increase in sales and customer satisfaction.

LEADERSHIP EXPERIENCE

Singapore Polytechnic International Students' Club (SPISC)

SINGAPORE

Secretary & Community Program Organizer of the SP International Students' Club

July 2021 – March 2024

(Key Skills: Multiple Stakeholders, Communication, Self & Team Management, Problem Solving, Creative Thinking)

- Fostered a fun work environment where CCA members across 8 departments can cooperate and achieve project milestones for all CCA events while tracking CCA matters and taking care of the members' welfare.
- Coordinated with the polytechnic administration, faculty, and staff to maintain smooth collaboration and participation in events, resulting in high turnout and engagement from students, with at least 150 student participants per event.

COMPETITION EXPERIENCE

Google Data Centre Outreach Hackathon

SINGAPORE

4th Place Winner - Team

October 2025

(Key Skills: Linux, Networking, Server Assembly, Troubleshooting, Communication)

- Directed a six-person team for rapid physical assembly of a disassembled enterprise server (fans, power supply, storage, and network cabling), leveraging prior PC-building expertise to accelerate the hardware phase and ensuring team success in the initial critical stage.
- Resolved complex server-to-VM networking issues and debugged unfamiliar server BIOS configurations under a strict time constraint, navigating enterprise hardware interfaces and applying methodical troubleshooting steps beyond standard PC systems.